

▼ More Important Functions

```
# value_counts
# sort_values
# rank
# sort index
# set index
# rename index -> rename
# reset index
# unique & nunique
# isnull/notnull/hasnans
# dropna
# fillna
# drop_duplicates
# drop
# apply
# isin
# corr
# nlargest -> nsmallest
# insert
# copy
```

```
import numpy as np
import pandas as pd
```

```
a = pd.Series([1,1,1,2,2,3])
a.value_counts()
```

```
↗ 1    3
   2    2
   3    1
dtype: int64
```

```
# value_counts(series and dataframe)
```

```
marks = pd.DataFrame([
    [100,80,10],
    [90,70,7],
    [120,100,14],
    [80,70,14],
    [80,70,14]
],columns=['iq','marks','package'])
```

```
marks
```

```
↗
```

	iq	marks	package
0	100	80	10
1	90	70	7
2	120	100	14
3	80	70	14
4	80	70	14

```
marks.value_counts()
```

```
↗
```

iq	marks	package	
80	70	14	2
90	70	7	1
100	80	10	1
120	100	14	1

dtype: int64

```
ipl = pd.read_csv('ipl-matches.csv')
ipl[~ipl['MatchNumber'].str.isdigit()]['Player_of_Match'].value_counts()
```

```
↗
```

KA Pollard	3
F du Plessis	3
SK Raina	3

```

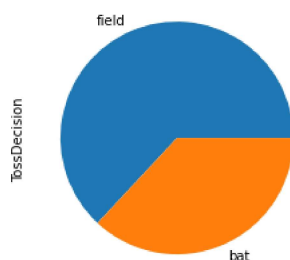
A Kumble                2
MK Pandey               2
YK Pathan               2
M Vijay                 2
JJ Bumrah               2
AB de Villiers          2
SR Watson               2
HH Pandya               1
Harbhajan Singh         1
A Nehra                 1
V Sehwag                1
UT Yadav                1
MS Bisla                1
BJ Hodge                1
MEK Hussey              1
MS Dhoni                1
CH Gayle                1
MM Patel                1
DE Bollinger            1
AC Gilchrist            1
RG Sharma               1
DA Warner               1
MC Henriques            1
JC Buttler              1
RM Patidar              1
DA Miller               1
VR Iyer                 1
SP Narine               1
RD Gaikwad              1
TA Boult                1
MP Stoinis              1
KS Williamson           1
RR Pant                 1
SA Yadav                1
Rashid Khan             1
AD Russell              1
KH Pandya               1
KV Sharma               1
NM Coulter-Nile         1
Washington Sundar       1
BCJ Cutting             1
M Ntini                 1
Name: Player_of_Match, dtype: int64

```

```
# find which player has won most potm -> in finals and qualifiers
```

```
# Toss decision plot
ipl['TossDecision'].value_counts().plot(kind='pie')
```

```
↗ <matplotlib.axes._subplots.AxesSubplot at 0x7f034efd49d0>
```



```
# how many matches each team has played
(ipl['Team2'].value_counts() + ipl['Team1'].value_counts()).sort_values(ascending=False)
```

```

↗ Mumbai Indians                231
  Royal Challengers Bangalore    226
  Kolkata Knight Riders          223
  Chennai Super Kings            208
  Rajasthan Royals               192
  Kings XI Punjab                190
  Delhi Daredevils               161
  Sunrisers Hyderabad            152
  Deccan Chargers                75
  Delhi Capitals                  63
  Pune Warriors                  46
  Gujarat Lions                  30
  Punjab Kings                   28
  Gujarat Titans                  16

```

```

Rising Pune Supergiant      16
Lucknow Super Giants        15
Kochi Tuskers Kerala        14
Rising Pune Supergiants     14
dtype: int64

```

```
# sort_values(series and dataframe) -> ascending -> na_position -> inplace -> multiple cols
```

```

x = pd.Series([12,14,1,56,89])
x

```

```

0    12
1    14
2     1
3    56
4    89
dtype: int64

```

```
x.sort_values(ascending=False)
```

```

4    89
3    56
1    14
0    12
2     1
dtype: int64

```

```

movies = pd.read_csv('movies.csv')
movies.head(4)

```

```

0    Uri: The Surgical Strike    tt8291224    https://upload.wikimedia.org/wikipedia/en/thum...    https://en.wikipedia.org/wiki/Uri:_The_Surgica...    Uri: The Surgical Strike    Uri: The Surgical Strike    is_a

1    Battalion 609    tt9472208    NaN    https://en.wikipedia.org/wiki/Battalion_609    Battalion 609    Battalion 609

2    The Accidental Prime Minister (film)    tt6986710    https://upload.wikimedia.org/wikipedia/en/thum...    https://en.wikipedia.org/wiki/The_Accidental_P...    The Accidental Prime Minister    The Accidental Prime Minister

3    Why Cheat India    tt8108208    https://upload.wikimedia.org/wikipedia/en/thum...    https://en.wikipedia.org/wiki/Why_Cheat_India    Why Cheat India    Why Cheat India

```

```
movies.sort_values('title_x',ascending=False)
```



	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
1623	Zubeidaa	tt0255713	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Zubeidaa	Zubeidaa	Zubeidaa
939	Zor Lagaa Ke...Haiya!	tt1479857	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Zor_Lagaa_Ke...H...	Zor Lagaa Ke... Haiya!	Zor Lagaa Ke... Haiya!
756	Zokkomon	tt1605790	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Zokkomon	Zokkomon	Zokkomon
670	Zindagi Tere Naam	tt2164702	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Zindagi_Tere_Naam	Zindagi Tere Naam	Zindagi Tere Naam
778	Zindagi Na Milegi Dobara	tt1562872	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Zindagi_Na_Mileg...	Zindagi Na Milegi Dobara	Zindagi Na Milegi Dobara
...	...	...	...	...	...	...
1039	1971 (2007 film)	tt0983990	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/1971_(2007_film)	1971	1971
723	1920: The Evil Returns	tt2222550	https://upload.wikimedia.org/wikipedia/en/e/e7...	https://en.wikipedia.org/wiki/1920:_The_Evil_R...	1920: Evil Returns	1920: Evil Returns
287	1920: London	tt5638500	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/1920_London	1920 London	1920 London

Start coding or generate with AI.

```
students = pd.DataFrame(  
    {  
        'name': ['nitish', 'ankit', 'rupesh', np.nan, 'mrityunjay', np.nan, 'rishabh', np.nan, 'aditya', np.nan],  
        'college': ['bit', 'iit', 'vit', np.nan, np.nan, 'vlsi', 'ssit', np.nan, np.nan, 'git'],  
        'branch': ['eee', 'it', 'cse', np.nan, 'me', 'ce', 'civ', 'cse', 'bio', np.nan],  
        'cgpa': [6.66, 8.25, 6.41, np.nan, 5.6, 9.0, 7.4, 10, 7.4, np.nan],  
        'package': [4, 5, 6, np.nan, 6, 7, 8, 9, np.nan, np.nan]  
    }  
)  
  
students
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
3	NaN	NaN	NaN	NaN	NaN
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0
7	NaN	NaN	cse	10.00	9.0
8	aditya	NaN	bio	7.40	NaN
9	NaN	git	NaN	NaN	NaN

```
students.sort_values('name',na_position='first',ascending=False,inplace=True)
```

```
students
```



	name	college	branch	cgpa	package
3	NaN	NaN	NaN	NaN	NaN
5	NaN	vlsi	ce	9.00	7.0
7	NaN	NaN	cse	10.00	9.0
9	NaN	git	NaN	NaN	NaN
2	rupesh	vit	cse	6.41	6.0
6	rishabh	ssit	civ	7.40	8.0
0	nitish	bit	eee	6.66	4.0
4	mrityunjay	NaN	me	5.60	6.0
1	ankit	iit	it	8.25	5.0
8	aditya	NaN	bio	7.40	NaN

```
movies.sort_values(['year_of_release','title_x'],ascending=[True,False])
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
1623	Zubeidaa	tt0255713	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Zubeidaa	Zubeidaa	Zubeidaa
1625	Yeh Zindagi Ka Safar	tt0298607	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Zindagi_Ka_S...	Yeh Zindagi Ka Safar	Yeh Zindagi Ka Safar
1622	Yeh Teraa Ghar Yeh Meraa Ghar	tt0298606	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Teraa_Ghar_Y...	Yeh Teraa Ghar Yeh Meraa Ghar	Yeh Teraa Ghar Yeh Meraa Ghar
1620	Yeh Raaste Hain Pyaar Ke	tt0292740	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Raaste_Hain_...	Yeh Raaste Hain Pyaar Ke	Yeh Raaste Hain Pyaar Ke
1573	Yaadein (2001 film)	tt0248617	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yaadein_(2001_film)	Yaadein...	Yaadein...
...	...	...	...	...	...	...
37	Article 15 (film)	tt10324144	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Article_15_(film)	Article 15	Article 15
46	Arjun Patiala	tt7881524	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Arjun_Patiala	Arjun Patiala	Arjun Patiala
10	Amavas	tt8396186	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Amavas	Amavas	Amavas
26	Albert Pinto Ko Gussa Kyun Aata Hai?	tt4355838	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Albert_Pinto_Ko_...	Albert Pinto Ko Gussa Kyun Aata Hai?	Albert Pinto Ko Gussa Kyun Aata Hai?

Start coding or [generate](#) with AI.

```
# rank(series)
batsman = pd.read_csv('batsman_runs IPL.csv')
batsman.head()
```

	batter	batsman_run
0	A Ashish Reddy	280
1	A Badoni	161
2	A Chandila	4
3	A Chopra	53
4	A Choudhary	25

```
batsman['batting_rank'] = batsman['batsman_run'].rank(ascending=False)
batsman.sort_values('batting_rank')
```

```

↵

```

	batter	batsman_run	batting_rank
569	V Kohli	6634	1.0
462	S Dhawan	6244	2.0
130	DA Warner	5883	3.0
430	RG Sharma	5881	4.0
493	SK Raina	5536	5.0
...	...	...	...
512	SS Cottrell	0	594.0
466	S Kaushik	0	594.0
203	IC Pandey	0	594.0
467	S Ladda	0	594.0
468	S Lamichhane	0	594.0

605 rows × 3 columns

```
# sort_index(series and dataframe)
```

```
marks = {
    'maths':67,
    'english':57,
    'science':89,
    'hindi':100
}
```

```
marks_series = pd.Series(marks)
marks_series
```

```

↵
maths      67
english    57
science    89
hindi     100
dtype: int64

```

```
marks_series.sort_index(ascending=False)
```

```

↵
science    89
maths      67
hindi     100
english    57
dtype: int64

```

```
movies.sort_index(ascending=False)
```

	title_x	imdb_id	poster_path	wiki_link	title_y	original_title
1628	Humsafar	tt2403201	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Humsafar	Humsafar	Humsafar
1627	Daaka	tt10833860	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Daaka	Daaka	Daaka
1626	Sabse Bada Sukh	tt0069204	NaN	https://en.wikipedia.org/wiki/Sabse_Bada_Sukh	Sabse Bada Sukh	Sabse Bada Sukh
1625	Yeh Zindagi Ka Safar	tt0298607	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Zindagi_Ka_S...	Yeh Zindagi Ka Safar	Yeh Zindagi Ka Safar
1624	Tera Mera Saath Rahen	tt0301250	https://upload.wikimedia.org/wikipedia/en/2/2b...	https://en.wikipedia.org/wiki/Tera_Mera_Saath_...	Tera Mera Saath Rahen	Tera Mera Saath Rahen
...	...	...	...	...	...	...
4	Evening Shadows	tt6028796	NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Cheat_India	Why Cheat India	Why Cheat India
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...	The Accidental Prime Minister	The Accidental Prime Minister
1	Battalion 609	tt9472208	NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609

```
# set_index(dataframe) -> inplace
batsman.set_index('batter', inplace=True)
```

```
batsman
```




	batsman_run	batting_rank
batter		
A Ashish Reddy	280	166.5
A Badoni	161	226.0
A Chandila	4	535.0
A Chopra	53	329.0
A Choudhary	25	402.5
...	...	...
Yash Dayal	0	594.0
Yashpal Singh	47	343.0
Younis Khan	3	547.5
Yuvraj Singh	2754	27.0
Z Khan	117	256.0

605 rows × 2 columns

```
# reset_index(series + dataframe) -> drop parameter
batsman.reset_index(inplace=True)
```



	batter	batsman_run	batting_rank
0	A Ashish Reddy	280	166.5
1	A Badoni	161	226.0
2	A Chandila	4	535.0
3	A Chopra	53	329.0
4	A Choudhary	25	402.5
...	...	...	...
600	Yash Dayal	0	594.0
601	Yashpal Singh	47	343.0
602	Younis Khan	3	547.5
603	Yuvraj Singh	2754	27.0
604	Z Khan	117	256.0

605 rows × 3 columns

batsman



	batsman_run	batting_rank
batter		
A Ashish Reddy	280	166.5
A Badoni	161	226.0
A Chandila	4	535.0
A Chopra	53	329.0
A Choudhary	25	402.5
...	...	...
Yash Dayal	0	594.0
Yashpal Singh	47	343.0
Younis Khan	3	547.5
Yuvraj Singh	2754	27.0
Z Khan	117	256.0

605 rows × 2 columns

```
# how to replace existing index without losing
batsman.reset_index().set_index('batting_rank')
```



	batter	batsman_run
batting_rank		
166.5	A Ashish Reddy	280
226.0	A Badoni	161
535.0	A Chandila	4
329.0	A Chopra	53
402.5	A Choudhary	25
...	...	...
594.0	Yash Dayal	0
343.0	Yashpal Singh	47
547.5	Younis Khan	3
27.0	Yuvraj Singh	2754
256.0	Z Khan	117

605 rows × 2 columns

```
# series to dataframe using reset_index
marks_series.reset_index()
```



	index	0
0	maths	67
1	english	57
2	science	89
3	hindi	100

```
# rename(dataframe) -> index
```

```
movies.set_index('title_x',inplace=True)
```

```
movies.rename(columns={'imdb_id':'imdb','poster_path':'link'},inplace=True)
```

```
movies.rename(index={'Uri: The Surgical Strike':'Uri','Battalion 609':'Battalion'})
```

		imdb	link	wiki_link	title_y	original_title	is
title_x							
Uri	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Uri:_The_Surgica...		Uri: The Surgical Strike	Uri: The Surgical Strike	
Battalion	tt9472208		NaN	https://en.wikipedia.org/wiki/Battalion_609	Battalion 609	Battalion 609	
The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/The_Accidental_P...		The Accidental Prime Minister	The Accidental Prime Minister	
Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Why_Cheat_India		Why Cheat India	Why Cheat India	
Evening Shadows	tt6028796		NaN	https://en.wikipedia.org/wiki/Evening_Shadows	Evening Shadows	Evening Shadows	
...	...		...		...	...	
Tera Mera Saath Rahen	tt0301250	https://upload.wikimedia.org/wikipedia/en/2/2b...	https://en.wikipedia.org/wiki/Tera_Mera_Saath_...		Tera Mera Saath Rahen	Tera Mera Saath Rahen	
Yeh Zindagi Ka Safar	tt0298607	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/wiki/Yeh_Zindagi_Ka_S...		Yeh Zindagi Ka Safar	Yeh Zindagi Ka Safar	
Sabse Bada Sukh	tt0069204		NaN	https://en.wikipedia.org/wiki/Sabse_Bada_Sukh	Sabse Bada Sukh	Sabse Bada Sukh	
Daaka	tt10833860	https://upload.wikimedia.org/wikipedia/en/thum...		https://en.wikipedia.org/wiki/Daaka	Daaka	Daaka	

```
# unique(series)
temp = pd.Series([1,1,2,2,3,3,4,4,5,5,np.nan,np.nan])
print(temp)
```

```
0    1.0
1    1.0
2    2.0
3    2.0
4    3.0
5    3.0
6    4.0
7    4.0
8    5.0
9    5.0
10   NaN
11   NaN
dtype: float64
```

```
len(temp.unique())
```

```
6
```

```
temp.nunique()
```

 5

```
len(ipl['Season'].unique())
```

 15

```
# nunique(series + dataframe) -> does not count nan -> dropna parameter
ipl['Season'].nunique()
```

 15

```
# isnull(series + dataframe)
students['name'][students['name'].isnull()]
```



```
3    NaN
5    NaN
7    NaN
9    NaN
Name: name, dtype: object
```

```
# notnull(series + dataframe)
students['name'][students['name'].notnull()]
```



```
0    nitish
1    ankit
2    rupesh
4    mrityunjay
6    rishabh
8    aditya
Name: name, dtype: object
```

```
# hasnans(series)
students['name'].hasnans
```

 True

```
students
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
3	NaN	NaN	NaN	NaN	NaN
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0
7	NaN	NaN	cse	10.00	9.0
8	aditya	NaN	bio	7.40	NaN
9	NaN	git	NaN	NaN	NaN

```
students.isnull()
```



	name	college	branch	cgpa	package
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	True	True	True	True	True
4	False	True	False	False	False
5	True	False	False	False	False
6	False	False	False	False	False
7	True	True	False	False	False
8	False	True	False	False	True
9	True	False	True	True	True

```
students.notnull()
```



	name	college	branch	cgpa	package
0	True	True	True	True	True
1	True	True	True	True	True
2	True	True	True	True	True
3	False	False	False	False	False
4	True	False	True	True	True
5	False	True	True	True	True
6	True	True	True	True	True
7	False	False	True	True	True
8	True	False	True	True	False
9	False	True	False	False	False

```
# dropna(series + dataframe) -> how parameter -> works like or
students['name'].dropna()
```



```
0      nitish
1      ankit
2      rupesh
4  mrityunjay
6      rishabh
8      aditya
Name: name, dtype: object
```

```
students
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
3	NaN	NaN	NaN	NaN	NaN
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0
7	NaN	NaN	cse	10.00	9.0
8	aditya	NaN	bio	7.40	NaN
9	NaN	git	NaN	NaN	NaN

```
students.dropna(how='any')
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
6	rishabh	ssit	civ	7.40	8.0

```
students.dropna(how='all')
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0
7	NaN	NaN	cse	10.00	9.0
8	aditya	NaN	bio	7.40	NaN
9	NaN	git	NaN	NaN	NaN

```
students.dropna(subset=['name'])
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
4	mrityunjay	NaN	me	5.60	6.0
6	rishabh	ssit	civ	7.40	8.0
8	aditya	NaN	bio	7.40	NaN

```
students.dropna(subset=['name', 'college'])
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
6	rishabh	ssit	civ	7.40	8.0

```
students.dropna(inplace=True)
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
3	NaN	NaN	NaN	NaN	NaN
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0
7	NaN	NaN	cse	10.00	9.0
8	aditya	NaN	bio	7.40	NaN
9	NaN	git	NaN	NaN	NaN

```
# fillna(series + dataframe)
students['name'].fillna('unknown')
```



```
0      nitish
1      ankit
2      rupesh
3      unknown
4  mrityunjay
5      unknown
6      rishabh
7      unknown
8      aditya
9      unknown
Name: name, dtype: object
```

```
students
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
3	NaN	NaN	NaN	NaN	NaN
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0
7	NaN	NaN	cse	10.00	9.0
8	aditya	NaN	bio	7.40	NaN
9	NaN	git	NaN	NaN	NaN

```
students['package'].fillna(students['package'].mean())
```



```
0      4.000000
1      5.000000
2      6.000000
3      6.428571
4      6.000000
5      7.000000
6      8.000000
7      9.000000
8      6.428571
9      6.428571
Name: package, dtype: float64
```

```
students['name'].fillna(method='bfill')
```



```
0      nitish
1      ankit
2      rupesh
3  mrityunjay
4  mrityunjay
```

```

5      rishabh
6      rishabh
7      aditya
8      aditya
9      NaN
Name: name, dtype: object

```

```
# drop_duplicates(series + dataframe) -> works like and -> duplicated()
```

```
temp = pd.Series([1,1,1,2,3,3,4,4])
temp.drop_duplicates()
```

```

0      1
3      2
4      3
6      4
dtype: int64

```

```
marks.drop_duplicates(keep='last')
```

```

iq  marks  package
0  100     80      10
1   90     70       7
2  120    100     14
4   80     70     14

```

```

# find the last match played by virat kohli in Delhi
ipl['all_players'] = ipl['Team1Players'] + ipl['Team2Players']
ipl.head()

```

	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	...	WinningTeam	WonBy
0	1312200	Ahmedabad	2022-05-29	2022	Final	Rajasthan Royals	Gujarat Titans	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	bat	...	Gujarat Titans	Wickets
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	Royal Challengers Bangalore	Rajasthan Royals	Narendra Modi Stadium, Ahmedabad	Rajasthan Royals	field	...	Rajasthan Royals	Wickets
2	1312198	Kolkata	2022-05-25	2022	Eliminator	Royal Challengers Bangalore	Lucknow Super Giants	Eden Gardens, Kolkata	Lucknow Super Giants	field	...	Royal Challengers Bangalore	Runs
3	1312197	Kolkata	2022-05-24	2022	Qualifier 1	Rajasthan Royals	Gujarat Titans	Eden Gardens, Kolkata	Gujarat Titans	field	...	Gujarat Titans	Wickets
4	1304116	Mumbai	2022-05-22	2022	70	Sunrisers Hyderabad	Punjab Kings	Wankhede Stadium, Mumbai	Sunrisers Hyderabad	bat	...	Punjab Kings	Wickets

5 rows × 21 columns

```

def did_kohli_play(players_list):
    return 'V Kohli' in players_list

```

```

ipl['did_kohli_play'] = ipl['all_players'].apply(did_kohli_play)
ipl[(ipl['City'] == 'Delhi') & (ipl['did_kohli_play'] == True)].drop_duplicates(subset=['City', 'did_kohli_play'], keep='first')

```




	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue	TossWinner	TossDecision	...	WonBy	Margin	method	Player_
208	1178421	Delhi	2019-04-28	2019	46	Delhi Capitals	Royal Challengers Bangalore	Arun Jaitley Stadium	Delhi Capitals	bat	...	Runs	16.0	NaN	

1 rows × 22 columns

```
students.drop_duplicates()
```

```
# drop(series + dataframe)
temp = pd.Series([10,2,3,16,45,78,10])
temp
```



```
0    10
1     2
2     3
3    16
4    45
5    78
6    10
dtype: int64
```

```
temp.drop(index=[0,6])
```



```
1     2
2     3
3    16
4    45
5    78
dtype: int64
```

```
students
```



	name	college	branch	cgpa	package
0	nitish	bit	eee	6.66	4.0
1	ankit	iit	it	8.25	5.0
2	rupesh	vit	cse	6.41	6.0
3	NaN	NaN	NaN	NaN	NaN
4	mrityunjay	NaN	me	5.60	6.0
5	NaN	vlsi	ce	9.00	7.0
6	rishabh	ssit	civ	7.40	8.0